

PROJECT PHASE – 3

1. Relational Model:

- **Mapping Multivalued Attribute (Umpires) to Relational Model:**

In the ER diagram, the attribute "**Umpires**" was a multivalued attribute, which violates the principles of the relational model since relational tables cannot have multivalued attributes. To address this, we created two separate relations:

- **Match_Umpires Relation:**

This relation captures the many-to-many relationship between matches and umpires. It has the following attributes:

- ✓ Match_ID: A foreign key referencing the match in the Match table.
- ✓ Umpire_ID: A foreign key referencing the umpire in the Umpires table.

The Match_Umpires table represents the assignments of multiple umpires to each match, removing the multivalued attribute from the original Match entity.

- **Umpires Relation:**

This relation stores the details of each umpire, including their first name, middle name, and last name:

- ✓ Umpire_ID: A unique identifier for each umpire (primary key).
- ✓ FName, MName, LName: Attributes representing the umpire's full name

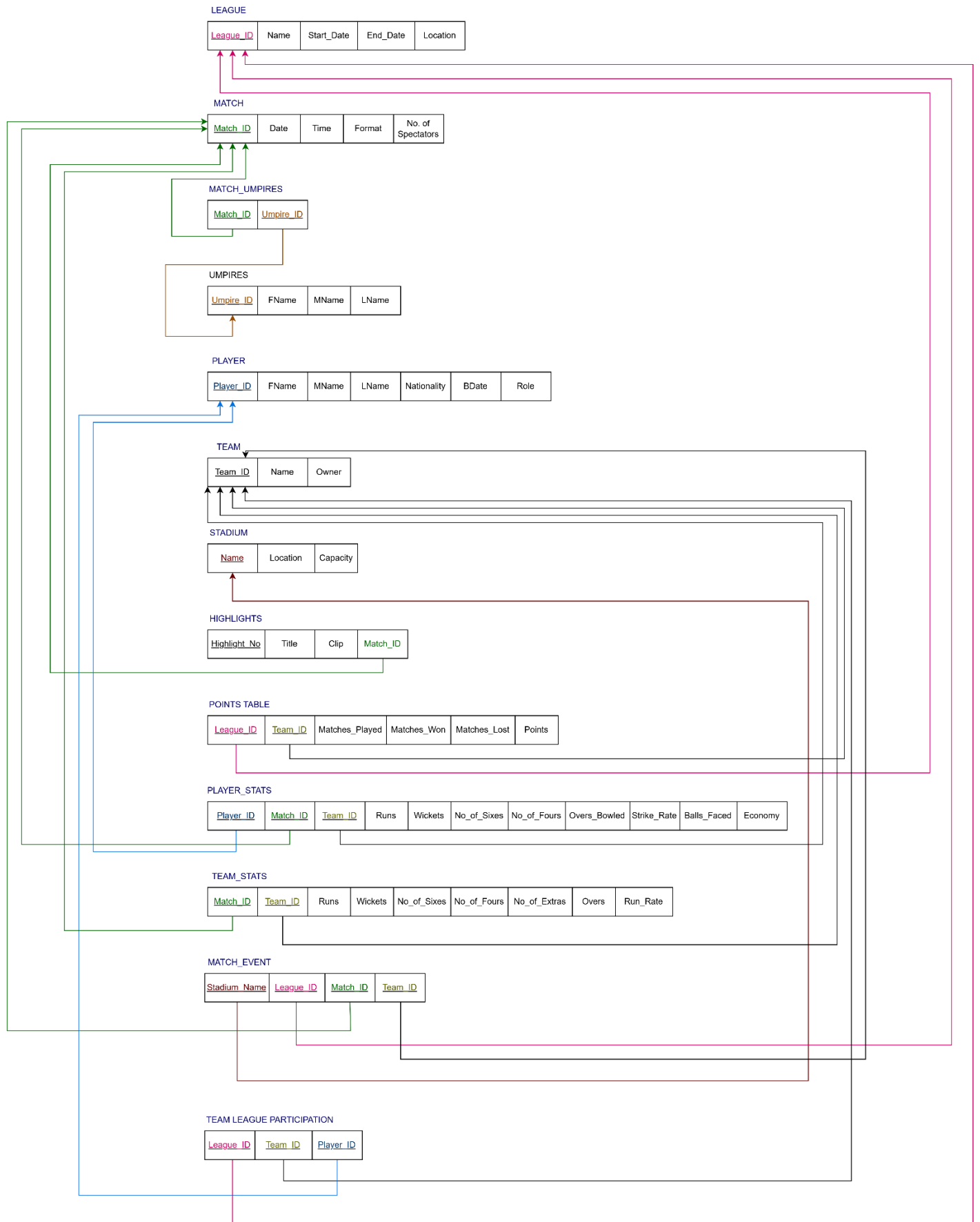
- **Handling Subclasses (Player_Stats, Team_Stats) in the Relational Model:**

In the ER diagram, we had a superclass **Scorecard** with two subclasses: **Player_Stats** and **Team_Stats**. To convert this structure to the relational model, we created separate relations for the subclasses (since the subclasses are disjoint and exhaustive):

- Player_Stats Relation

- Team_Stats Relation

SNAPSHOT OF RELATIONAL MODEL



2. First Normal Form(1NF):

Our relational model already adhered to **1NF** as all attributes contain atomic values and there are no repeating groups or multivalued attributes. So, no changes were required to be made to the relational model to convert it into the 1NF Form.

3. Second Normal Form(2NF):

- The changes made to the relational model to convert it to the Second Normal Form are as follows:

Match_Event Relation:

In the original Match_Event relation, the attribute Stadium_Name exhibited a partial dependency on the Match_ID attribute. So, to satisfy the constraints of 2NF, this relation has been broken down into two separate relations **Match_Event** and **Match_Location**.

- ✓ Match_Event :
Attributes : League_ID, Match_ID, Team_ID
- ✓ Match_Location:
Attributes: Match_Id, Stadium_Name

Removal of the attribute Team_ID from the relation Player_Stats:

The attribute was removed as it had partial dependency with Match_ID. However that dependency between Match_ID and Team_ID is already addressed in the relation Match_Event.

4. Third Normal Form(3NF):

After converting to 2NF, we reviewed the model and found no further transitive dependencies. the 2NF model already satisfies **Third Normal Form (3NF) constraints**, and no further changes were necessary.

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LEAGUE				
League_ID	Name	Start_Date	End_Date	Location

MATCH					
Match_ID	Date	Time	Format	No. of Spectators	

MATCH_UMPIRES	
Match_ID	Umpire_ID

UMPIRES			
Umpire_ID	FName	MName	LName

PLAYER						
Player_ID	FName	MName	LName	Nationality	BDate	Role

TEAM		
Team_ID	Name	Owner

STADIUM		
Name	Location	Capacity

HIGHLIGHTS			
Highlight_No	Title	Clip	Match_ID

POINTS TABLE					
League_ID	Team_ID	Matches_Played	Matches_Won	Matches_Lost	Points

PLAYER_STATS									
Player_ID	Match_ID	Runs	Wickets	No_of_Sixes	No_of_Fours	Overs_Bowled	Strike_Rate	Balls_Faced	Economy

TEAM_STATS								
Match_ID	Team_ID	Runs	Wickets	No_of_Sixes	No_of_Fours	No_of_Extras	Overs	Run_Rate

MATCH_EVENT		
League_ID	Match_ID	Team_ID

MATCH_LOCATION	
Match_ID	Stadium_Name

TEAM_LEAGUE_PARTICIPATION		
Team_ID	Player_ID	League_ID

